Piperacillin with Tazobactam

FORM Vials containing: 2.25g (piperacillin 2g and tazobactam 250mg) or 4.5g (piperacillin 4g and tazobactam 500mg) powder for reconstitution

INDICATION Treatment of infection where guidelines or sensitivities indicate

DOSE RANGE

AGE	DOSE	FREQUENCY	ROUTE
Birth – 1month	90mg/kg*	Every 8 hours	IV
> 1 month	90mg/kg*	Every 6- 8 hours	IV

- *Doses stated are the combined drug content. Each 90mg is equivalent to Piperacillin 80mg and Tazobactam 10mg.

For positive cultures where the isolate is reported as 'I' an increased exposure dosing regimen is required. Exposure may be increased as a function of dose, frequency or administration time, depending on the drug.
Please refer to the following GGC RHC guidance "Isolates reported as 'I' (susceptible increased exposure) - dosing schedule for neonates and children" or contact Pharmacy for further information.

RECONSTITUTION

2.25g vial: add 8.4ml of either water for injection or sodium chloride 0.9% to give a final concentration of 225mg/ml

4.5g vial: add 16.8ml of either water for injection or sodium chloride 0.9% to give a final concentration of 225mg/ml

Vials should be swirled until the powder is fully dissolved. Reconstitution may take 5-10 minutes for some products.

FURTHER DILUTION :

Further dilution is not required for most infants.
90mg/kg = 0.4ml/kg of reconstituted solution (225mg/ml)

For ELBW infants,

Piperacillin/tazobactam 225mg/ml	2ml
Compatible infusion fluid	Up to 5ml total

Gives a 90mg in 1ml solution. Use the required volume.

METHOD OF ADMINISTRATION

Slow IV injection over 3-5minutes

COMPATIBILITY

Solution compatibility	sodium chloride 0.45%, sodium chloride 0.9%, glucose 5%, Water for Injection
Solution incompatibility	Hartmann's solution
IV line compatibility	Dopamine,dexamethasone, dexmedatomidine,fluconazole, furosemide, hydrocortisone sodium succinate, linezolid, metronidazole, milrinone, morphine, paracetamol, potassium chloride
IV line incompatibility	Aciclovir, amikacin, amphotericin B (conventional and liposomal), caspofungin, ciprofloxacin, dobutamine, ganciclovir, gentamicin, tobramycin, vancomycin

THIS LIST IS NOT EXHAUSTIVE PLEASE CONTACT PHARMACY FOR FURTHER INFORMATION ON COMPATIBILITY WITH ANY MEDICINES NOT INCLUDED

CAUTIONS, CONTRA-INDICATIONS AND SIDE EFFECTS

- See Summary of Product Characteristics and most recent edition of BNF for Children

FURTHER INFORMATION

Reduce dose in renal impairment – see BNF-C for dosing advice. May cause neurological toxicity such as seizures if accumulation occurs

Avoid mixing in the giving set with aminoglycosides - flush line between doses or administer at separate sites.

May prolong neuromuscular blockade of vecuronium and other non-depolarising muscle relaxants

Piperacilln/tazobactam has a high osmolarity – monitor cannula site with peripheral administration.

Hypokalaemia, hypoglycaemia and hypotension may occur

SODIUM CONTENT Average 4.7mmol per 2.25g

PH 4.5 – 7.0

LICENSED STATUS Not licensed for use in children < 2 years

APPLICABLE POLICIES West of Scotland Neonatal Guidelines:

Consult local policy if applicable

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Administer reconstituted solutions immediately.
All vials, ampoules and infusion bags are for single use only unless otherwise stated.

Dose may vary depending on indication, age, renal function, hepatic function, and concomitant medications.
This monograph should be used in conjunction with the terms of reference document prepared by the West of Scotland Pharmacist Group. Information is correct at the time of publication and as per local practice agreement.
For further advice please contact your clinical pharmacist or pharmacy department